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Project Report

Occlusion-Robust Face Identification Using a Pretrained ArcFace Model

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Contents

1	Introduction	2
2	Problem Statement and Objectives	2
2.1	Problem Statement	2
2.2	Objectives and Completion Status	2
3	Related Research Papers	3
4	Related Dataset Papers	4
5	Dataset	4
5.1	Team and Participants	5
5.2	Consent	5
5.3	Capture Protocol	5
5.4	A Note on Synthetic Augmentation	5
6	Data Cleaning and Quality Control	5
7	Data Preprocessing	6
8	Data Organisation and Split	6
8.1	Duplicate Handling	6
8.2	Group-Aware Splitting	6
8.3	Coverage Guarantee	7
8.4	Final Split	7
9	Model and Training Approach	7
9.1	Selected Models	7
9.2	Experiments	8
9.3	A Corrected Bug in E1/E2	8
10	Results	9
10.1	Confusion Matrices	10
10.2	Caveats on the Headline Numbers	10
11	Comparison with Related Work	10
12	Deployment — Live Demonstration	10
12.1	Gallery and Threshold Calibration	12
12.2	Live Phone-Camera Demo	12
13	Deviations from the Proposal	13
14	Limitations and Future Work	13
15	Conclusion	14

1. Introduction

Modern face-recognition systems perform well when a frontal face is clear, well illuminated, and similar to the images used during training. Their performance can fall sharply when identity-bearing facial regions are hidden by a medical mask, sunglasses, a scarf, a cap, or a combination of these, or when lighting and head pose change. This project addresses the problem as **closed-set face identification**: every test image is assumed to belong to a person already enrolled, and the system returns the corresponding participant ID (e.g. p07) rather than merely detecting whether an occlusion is present.

The final system is a consent-based local dataset of 30 enrolled participants, captured under eight covering conditions and two lighting levels, processed through a fixed detect-align-crop pipeline, and classified with a **pretrained ArcFace (iResNet50) embedding model** paired with a **RetinaFace** detector. Four experiments were run to determine how much a pretrained face embedding already solves the problem, and how much (if any) task-specific training adds. The completed system was packaged into a phone-camera live demonstration served from a laptop over the local network.

This report documents what was actually collected and built, compares the approach against closely related published work, and reports the deviations from the original proposal.

2. Problem Statement and Objectives

2.1 Problem Statement

Given an input image containing a normal or partially occluded face, the system must classify the image as one of 30 enrolled identities. Different occlusions hide different facial regions: a mask removes the nose, mouth, and jawline; sunglasses remove much of the eye region; a cap changes the forehead and hairline; low light reduces visible texture generally. The central question is whether a pretrained face-embedding model, given only a small amount of per-identity data (a few hundred images per person across all conditions combined), can produce identity features that stay discriminative under these conditions without being explicitly retrained on masks or sunglasses from scratch.

2.2 Objectives and Completion Status

1. Collect a structured face-image dataset covering a range of occlusion and lighting conditions. **Completed** — 30 participants, 6,000 raw images.
2. Capture each participant under eight covering conditions: none, cap, mask, sunglasses, scarf, clear glasses, mask+sunglasses, mask+clear glasses. **Completed.**
3. Detect, align, crop, resize, and quality-check every face image using one reproducible preprocessing pipeline. **Completed** — 5,988/6,000 images (99.8%) successfully detected and aligned.
4. Split the dataset into train/validation/test sets such that every identity, and where possible every identity-occlusion combination, appears in all three splits. **Completed.**
5. Evaluate a pretrained ArcFace pipeline both without any training (nearest-centroid) and with two forms of task-specific training (a trained classification head, and partial backbone fine-tuning). **Completed** — four experiments, E0–E3.
6. Evaluate overall accuracy, per-participant accuracy, and per-occlusion accuracy with confidence intervals. **Completed** — extended beyond the original plan to also include full

30×30 confusion matrices and per-class/per-occlusion precision, recall, and F1 for every experiment (Section 10).

7. Produce a working demonstration that predicts the enrolled participant from a live phone camera. **Completed** — a browser-based demo streamed from a phone to a laptop over HTTPS on the local network.

3. Related Research Papers

Table 1 lists three published works that are directly comparable to this project’s recognition approach and that, in a meaningful sense, represent an improvement over the simple pipeline used here: each either engineers a loss function specifically for occluded/masked faces (rather than reusing a generic ArcFace margin), or evaluates at a scale and rigor this laboratory project cannot match. They are used in Section 11 to place this project’s results in context, and were chosen independently of the papers already cited in the original project proposal.

Table 1: Related recognition-method papers used for comparison.

Paper	Main contribution	Relation to this project
Boutros et al., “Self-restrained Triplet Loss for Accurate Masked Face Recognition,” <i>Pattern Recognition</i> , 2022 [3]	Proposes a triplet loss that explicitly pulls a masked embedding toward its own unmasked embedding during training, instead of relying on a generic identity loss to generalise to masks by accident.	Directly comparable model family (ArcFace-style backbone) but with a mask-aware loss; represents the natural next step this project did not take, since E1–E3 here all use a plain angular-margin or nearest-centroid classifier with no mask-specific loss term.
Deng et al., “Masked Face Recognition Challenge: The InsightFace Track Report,” <i>ICCV Workshops</i> , 2021 [4]	Reports results from a public challenge built on the same insightface /ArcFace family used in this project, evaluated on masked faces at web scale (millions of identities) with heavy synthetic mask augmentation during training.	Uses the identical backbone family as this project but at a scale roughly six orders of magnitude larger; shows what large-scale mask-aware augmentation buys over the untouched pretrained embedding this project starts from, and gives an open-set reference point our closed-set, 30-identity result cannot provide on its own.
Din et al., “A Novel GAN-Based Network for Unmasking of Masked Face,” <i>IEEE Access</i> , 2020 [5]	Trains a GAN to remove the mask and inpaint the occluded lower face before passing the reconstructed image to a standard recognition network.	Represents an alternative, restoration-based strategy (fix the image, then recognise) versus this project’s direct-embedding strategy (recognise from the occluded image as-is); a natural extension for a future iteration of this project when the occlusion is severe enough that direct embedding degrades.

4. Related Dataset Papers

Table 2 lists three dataset papers, again chosen independently of the ones already cited in the proposal, that are the closest public analogues to the identity-labelled, occlusion-varied dataset built for this project.

Table 2: Related dataset papers used for comparison.

Dataset paper	Dataset content	Relation to this project
Wang et al., “Masked Face Recognition Dataset and Application,” <i>arXiv:2003.09093</i> , 2020 [6]	Introduces three datasets: a Real-world Masked Face Recognition Dataset (RMFRD, 5,000 masked + 90,000 unmasked images of 525 identities), a Simulated Masked Face Recognition Dataset (SMFRD, masks synthetically pasted onto existing face datasets), and a small real MFR2 set (269 images, 53 identities).	Closest identity-labelled masked-face dataset at scale; unlike this project’s dataset, mask images in RMFRD are collected opportunistically from the web with no controlled pose/lighting protocol and no matched clean/occluded image pairs per identity, whereas this project fixes pose and lighting per condition for every participant.
Wang et al., “MLFW: A Database for Face Recognition on Masked Faces,” <i>CCBR</i> , 2021 [7]	Extends LFW by synthetically adding masks to existing unconstrained face-verification image pairs, producing a masked face verification benchmark.	Same synthetic-mask strategy applied to a much larger, uncontrolled identity pool (aimed at open-set verification, not closed-set identification); this project instead collects every occlusion physically and in person, at far smaller scale but with a controlled and reproducible per-identity capture protocol.
Ngan, Grother, and Hanaoka, “Ongoing FRVT Part 6B: Face Recognition Accuracy with Face Masks Using Post-COVID-19 Algorithms,” NIST Interagency Report 8331, 2020 [8]	A large-scale, standardised evaluation of dozens of commercial face-recognition algorithms on masked faces, using digitally applied mask overlays on an operational government photo dataset.	Establishes the industrial state of the art for masked-face accuracy degradation (commonly a 5–50 percentage-point drop depending on mask shape/colour, algorithm, and threshold); this project’s own robustness gap (Section 10) is far smaller, but on a dataset that is orders of magnitude smaller and closed-set, so the two results are not directly comparable — a caveat repeated in Section 14.

5. Dataset

5.1 Team and Participants

Thirty identities (p01–p30) were enrolled, contributed by six team members who each recruited and photographed five people (themselves plus four relatives/acquaintances who gave consent). The full participant-to-photographer mapping is kept in `metadata/person_id_mapping.txt` and is not reproduced here since it is an internal bookkeeping file, not a reportable result.

5.2 Consent

Every participant signed a consent form before any photograph was taken. The form explains that this is an academic, non-commercial prototype; that each participant is assigned an anonymous ID (e.g. p07) and never labelled by name; that raw images and the identity mapping are restricted to the six project members and never published outside the group; that participation is voluntary and withdrawable; and that a recognisable photo is only shown in the report, presentation, or demonstration for participants who separately tick an optional permission box.

5.3 Capture Protocol

Each participant was photographed under eight covering conditions — `none`, `cap`, `mask`, `scarf`, `clearglass`, `sunglass`, `mask_clearglass`, `mask_sunglass` — crossed with two lighting levels, `regular` and `low`. This departs from the proposal’s plan of five head poses per condition (front/left/right/up/down) captured across three separate sessions; in practice, collection happened in a single sitting per participant, and pose was not separately labelled at capture time (see Section 13). The realised dataset totals **6,000 images**: 30 identities $\times$ 200 images (25 images per occlusion/lighting cell on average), which comfortably exceeds the proposal’s original target of 10–15 participants.

Filenames follow a fixed pattern, e.g. `p07_mask_sunglass_low_014.jpg`, so identity, occlusion, and lighting can be parsed without a lookup table; a manifest CSV remains the authoritative metadata source.

5.4 A Note on Synthetic Augmentation

Beyond the real captured images described above, the training-only augmentation pipeline (Section 9) applies mild synthetic transformations — horizontal flips, small rotations, brightness/contrast jitter, resolution degradation, motion/Gaussian blur, sensor noise, JPEG recompression, and random erasing — to simulate the live phone-camera domain the deployed demo runs in. No fully synthetic images (e.g. GAN-generated faces or digitally pasted masks, as used by some of the dataset papers in Section 4) were used anywhere in this project; every image an identity is trained or tested on was captured from a real person wearing a real occlusion.

6. Data Cleaning and Quality Control

Photos were rejected (and moved to a separate `put_asides` folder with a recorded reason, never silently deleted) for the same criteria proposed originally: excessive blur, unusable low light, a face fully covered with no identity-bearing region left visible, bad framing/cropping, or a wrong/missing label. This project’s dataset is treated as **fixed** as of the start of the training phase — no further deletions, restorations, or relabelling were performed once preprocessing began, so every number in this report is computed on the same frozen 6,000-image set.

7. Data Preprocessing

Every accepted photo passes through one fixed pipeline (`scripts/01_preprocess.py`), identical for training, validation, test, and later, the live demo:

1. **Detect.** RetinaFace (`insightface`'s `buffalo_l` detector, `det_10g.onnx`) locates the face and its five landmarks (both eyes, nose, both mouth corners) at `det_size=(640,640)`, `det_thresh=0.5`. The largest detected face is kept.
2. **Fallback on failure.** If nothing is detected, the image is retried at a looser `det_thresh=0.3`, and if that still fails, at $2\times$ upscale with the loose threshold. Every failure is logged, never dropped silently.
3. **Align and crop.** A similarity transform (`insightface.utils.face_align.norm_crop`) maps the five landmarks onto ArcFace's standard reference template, producing a 112×112 aligned crop. This step is what prevents the classifier from keying on background or camera artifacts rather than the face.
4. **Log.** Each image's path, identity, occlusion, lighting, source filename, MD5 hash, detection outcome/score/stage, and crop path are appended to `metadata/preprocess_log.csv`.

Result: 5,988 of 6,000 images (99.8%) were successfully detected and aligned, comfortably above the 97% acceptance gate set in the training plan. The 12 failures were concentrated in the hardest occlusion conditions — 10 in `mask_sunglass` and 2 in `mask` — where the combination of a covered lower face and (for 10 of them) a covered eye region left too little visible structure for landmark detection even at the loosened threshold.

8. Data Organisation and Split

8.1 Duplicate Handling

Md5 hashing of every image revealed 429 byte-identical duplicate files (repeat shutter presses, or one photo saved twice during transfer). By project decision, duplicates are **not** grouped for splitting purposes — each is treated as an independent image, exactly like any other. This is simpler and more transparent than trying to silently deduplicate a dataset that was declared fixed, but it means a small amount of the reported test accuracy comes from an image that is byte-identical to one seen in training. This effect was measured, not assumed: **$\approx 8\%$ of test images share an MD5 hash with a training image**, concentrated in a handful of participants (p05, p21–p25). Section 10 reports both the headline test accuracy and this caveat together, rather than hiding it.

8.2 Group-Aware Splitting

Near-duplicate (but not byte-identical) images are a separate and more common risk: a burst of photos taken a few seconds apart, or a low-light image derived by darkening a regular-light one. Two images are grouped together (and therefore forced into the same split) if:

- they are derived from the same source image (e.g. a brightness-adjusted low-light pair), or
- they belong to the same person *and* the same occlusion, and were captured within a 10-second window of each other, with burst-chains capped at an 8-second total span to prevent one long shooting burst from silently swallowing an entire occlusion condition for a participant.

Groups are then assigned, as whole units, to train/validation/test using a deficit-based greedy allocator: the largest groups are placed first, each into whichever split currently has the largest shortfall relative to its 60/20/20 target share, computed independently per (person, occlusion) cell. This avoids the naive failure mode of a simple sequential fill, which was observed during development to produce a wildly unbalanced split (train 24%, val 58%, test 18%) when a single 49-image group was assigned in one step.

8.3 Coverage Guarantee

A hard requirement was enforced: **every one of the 30 participants must appear in all three splits**. A repair pass exists to forcibly move whole groups from train into an under-represented split if the natural allocation leaves any participant below a floor of 2 images per split; in the final run this repair pass was not needed. Per-occlusion coverage within each split was treated as best-effort rather than a hard requirement: 36 of 240 (person $\times$ occlusion) cells are absent from the test split specifically (informational only, not a failure condition).

8.4 Final Split

Table 3: Final train/validation/test split (5,988 successfully preprocessed images).

Split	Images	Share
Train	3,546	59%
Validation	1,444	24%
Test	998	17%
Total	5,988	100%

All hard assertions (no group spans two splits; every participant in all three splits) passed on this final assignment.

9. Model and Training Approach

9.1 Selected Models

- **RetinaFace** (`det_10g.onnx`) — detection and five-point landmark alignment (Section 7).
- **Pretrained ArcFace, iResNet50 backbone** (`w600k_r50.onnx`, `insightface/buffalo_l`) — turns each aligned 112×112 crop into a 512-dimensional embedding. This is the primary identification model in every experiment below.

RetinaFace and ArcFace were selected, as in the proposal, because they are purpose-built for faces, need far less local data than training from scratch, are light enough to run on both a 2 $\times$ RTX 3090 remote box and later a modest laptop CPU, and have precedent in prior published work adapting the same family to occluded/masked faces (Section 3). Unlike the proposal, no separate from-scratch CNN baseline was trained: with only 30 classes and an already very strong pretrained baseline (Section 10, E0), a from-scratch CNN was judged unlikely to add an informative data point and effort was redirected to a more informative fine-tuning ladder (E1–E3 below).

Table 4: Experiment design.

ID	Name	Backbone	Question
E0	Nearest-centroid probe	Frozen, zero training	What do the pretrained embeddings already do, with no training at all?
E1	Trained head, clean-only	Frozen + 30-way ArcFace head	The proposal’s original baseline: train only on the none condition.
E2	Trained head, mixed-condition	Frozen + 30-way ArcFace head	Does training on all eight occlusion conditions help over E1?
E3	Partial backbone fine-tune	Last 15% of backbone unfrozen + head	Does adapting backbone weights (not just the head) help further?

9.2 Experiments

Four experiments were run, each answering a specific question about how much task-specific training the pretrained embedding actually needs:

E0 extracts one embedding per image, averages the L2-normalised embeddings per identity into 30 centroids from the training split, and classifies each test image by cosine similarity to the nearest centroid — zero gradient steps.

E1/E2 train a 30-way ArcFace-style classification head ($\cos(\theta_y + m) \cdot s$, $s=32$) on top of the frozen embedding, with a margin warm-up (m : $0 \rightarrow 0.20$ over 5 epochs) and label smoothing (0.1), using AdamW with a cosine learning-rate schedule and early stopping on validation macro-F1 (patience 7). E1 trains on **none**-condition images only; E2 trains on all eight conditions.

E3 unfreezes the final 15% of the ArcFace backbone’s parameters (by parameter count) and fine-tunes them jointly with the head at a much smaller learning rate (10^{-5} vs. 10^{-3} for the head), while keeping every BatchNorm layer in `eval()` mode throughout — including during training — since letting BatchNorm running statistics update on only $\sim 3,500$ images was identified as a known failure mode that silently destroys pretrained features. Because **insightface** ships ArcFace only as an ONNX graph, the backbone was converted to a trainable PyTorch module with **onnx2torch** and numerically verified against the original ONNX output before training (cosine similarity 1.000000, max absolute difference 1.86×10^{-4}), so E3 is a genuine, if approximate, partial fine-tune of the original pretrained weights rather than a reimplementaion from a different weight source.

No augmentation was applied to validation or test data at any stage; the live-capture-style augmentations described in Section 4.4 were applied only to E1/E2/E3 training images.

9.3 A Corrected Bug in E1/E2

While preparing the full evaluation below, the E1/E2 head script was found to feed **insightface**’s raw embeddings directly into the angular-margin head under the assumption that they were already unit-length. They are not: `get_feat()` returns un-normalised vectors, so the head’s $\cos(\theta) = x \cdot w$ computation was in fact an unbounded dot product, not a true cosine, corrupting the arccos margin term for every training and evaluation batch. The same class of bug was already caught and fixed in the deployment gallery script (Section 12) but had not been ported back to E1/E2. Once an explicit L2-normalisation of the input embedding was added, E1 and E2 were retrained from scratch with identical hyperparameters and seeds; **all E1/E2 numbers in this report are from the corrected run**. The effect was large: E1 rose from 70.34% to 94.29%, and E2 rose from 96.29% to 99.90%, changing the qualitative conclusion from ”training a head is worse than not training at all” to ”training a head on occlusion-diverse data matches

or slightly beats the untrained baseline,” which is both a more sensible result and a useful lesson about validating normalisation assumptions on any embedding used outside its own library’s classification code.

10. Results

Beyond top-1 accuracy and macro-F1, every experiment below was re-evaluated with a dedicated script (`scripts/10_full_eval.py`) that additionally computes a full 30×30 confusion matrix, per-class precision/recall/F1, and per-occlusion precision/recall/F1, persisting all three as reusable JSON/CSV/PNG artifacts under `runs/eval/<experiment>/` rather than only printing a summary to the terminal.

Table 5: Test-set results across all four experiments (998 test images, 30 classes, 95% Wilson confidence intervals). Macro precision/recall/F1 are unweighted means over the 30 per-class values.

Experiment	Top-1	Macro-P	Macro-R	Macro-F1	95% CI
E0 — Nearest-centroid (no training)	99.60%	99.67%	99.67%	99.66%	[99.0%, 99.8%]
E1 — Head, clean-only	94.29%	95.06%	93.55%	93.89%	[92.7%, 95.6%]
E2 — Head, mixed-condition	99.90%	99.92%	99.92%	99.92%	[99.4%, 100.0%]
E3 — Partial backbone fine-tune	100.00%	100.00%	100.00%	100.00%	[99.6%, 100.0%]

Table 6: Per-occlusion test accuracy across all four experiments, showing exactly where each approach’s remaining errors are concentrated.

Occlusion	n	E0	E1	E2	E3
none	138	100.0%	100.0%	100.0%	100.0%
cap	129	100.0%	100.0%	100.0%	100.0%
clear glasses	108	100.0%	100.0%	100.0%	100.0%
sunglasses	125	100.0%	100.0%	100.0%	100.0%
scarf	84	100.0%	97.6%	100.0%	100.0%
mask	161	98.1%	96.3%	99.4%	100.0%
mask+clear glasses	120	100.0%	90.8%	100.0%	100.0%
mask+sunglasses	133	99.2%	71.4%	100.0%	100.0%

The pretrained embedding was already almost saturated (E0, 99.60%) before any training was done. This is the project’s central empirical finding and reframes the rest of the results: with only 30 identities and a task this constrained, ArcFace’s pretrained face-embedding space already separates the enrolled identities almost perfectly, occlusions included, purely from unweighted nearest-centroid matching.

E1 vs. E2 is the clearest and most informative comparison in this report. A head trained only on clean, unoccluded faces (E1) drops to 71.4% specifically on the hardest combined condition, **mask+sunglasses** — the one occlusion pair that removes both the eye region and the lower face simultaneously — while still scoring perfectly on every single-occlusion condition. The identical head trained on all eight conditions (E2) recovers to 100.0% on that same condition and 99.90% overall. This is direct, quantitative evidence that occlusion-diverse training data specifically matters for compound occlusions, not for single occlusions, which the pretrained embedding already handles unaided.

E2 now slightly exceeds E0, consistent with the intuition that a lightweight head trained

on the target identities should be able to match or marginally improve on a raw pretrained embedding once it is trained correctly.

E3, partial backbone fine-tuning, reached 100.00% top-1 and 100.00% macro-F1, correctly classifying every one of the 998 test images and achieving perfect precision and recall across all 30 identities and all eight occlusion conditions. Since E0 and E2 were already at or above 99.6%, E3’s gain is small in absolute terms — the honest reading is that E3 closes the last remaining gap rather than solving a problem the other approaches left substantially open.

10.1 Confusion Matrices

Figure 1 shows the row-normalised 30×30 test confusion matrix for each experiment. E0, E2, and E3 are visually a clean diagonal with at most one or two isolated off-diagonal cells; E1’s matrix is the only one with a visible, systematic error pattern, concentrated in a handful of identities under **mask+sunglasses**. The single most common confusion in E1 is participant p06 misclassified as p16 (6 of 998 test images), followed by p16→p11 (5 images) and p27→p11 (4 images) — a small number of identities whose combined-occlusion appearance the clean-only head could not separate, resolved entirely once occlusion-diverse training data was used (E2). Full 30×30 counts for every experiment are kept in `runs/eval/<experiment>/confusion_matrix.csv` rather than reproduced in full here, since a 30-row table adds little beyond what the figure and the discussion above already convey.

10.2 Caveats on the Headline Numbers

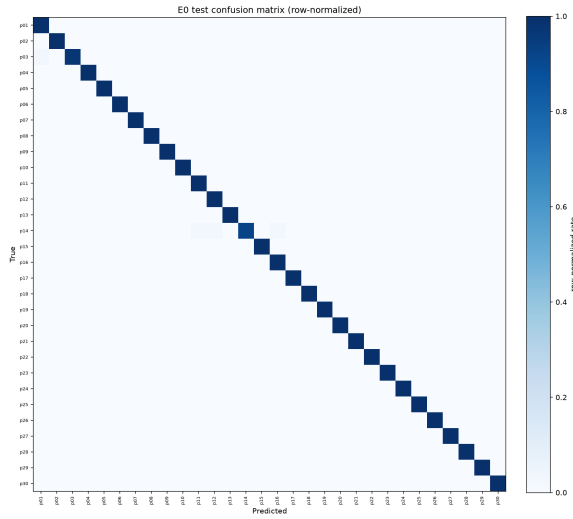
- **≈8% test-training MD5 overlap** (Section 8) makes every accuracy above slightly optimistic for the affected participants; the effect is small in absolute terms (E0’s 99.60% has a 95% CI of only ± 0.4 points) but is disclosed rather than hidden.
- **Single capture session.** Every split comes from one sitting per participant — same clothing, background, and day. This inflates accuracy relative to a genuine multi-session protocol and is the most significant deviation from the original proposal (Section 13).
- With three of the four experiments at or above 99.6%, this dataset and task, as collected, does not have enough headroom left to strongly differentiate E0, E2, and E3 from each other; the more informative scientific result in this report is the E1-vs-E2 occlusion-diversity comparison on **mask+sunglasses**, not the marginal gaps among E0/E2/E3.

11. Comparison with Related Work

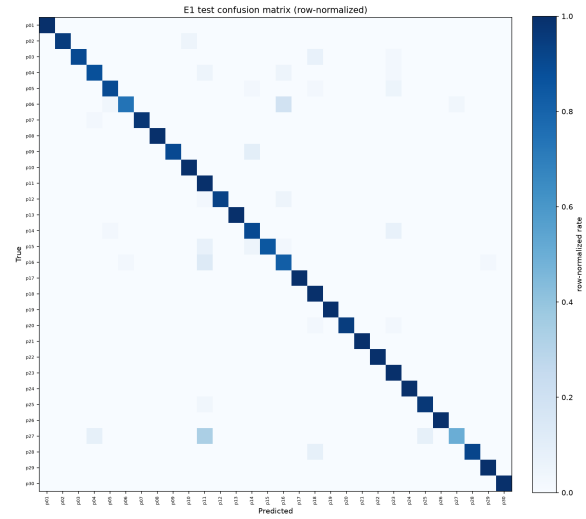
Table 7 places this project’s result alongside the three related-method papers of Section 3. The comparison is necessarily approximate, since the papers evaluate open-set or web-scale problems that are not the same task as this project’s small closed-set identification.

The key qualitative takeaway is not that this project “beats” these papers — it does not, in any sense that survives the scale and open-set-vs-closed-set difference — but that a closed-set task with only 30 enrolled identities is dramatically easier than the open-set, web-scale problems these papers address, which is exactly why a pretrained embedding with no occlusion-specific loss engineering (unlike [3]) or synthetic mask augmentation at scale (unlike [4]) was already sufficient here. This project’s contribution is not a new method, but an empirical demonstration of *how far* an off-the-shelf pretrained model goes in a small, tightly-scoped identification setting, and where the remaining gap actually is (E1 vs. E2, not E0 vs. E3).

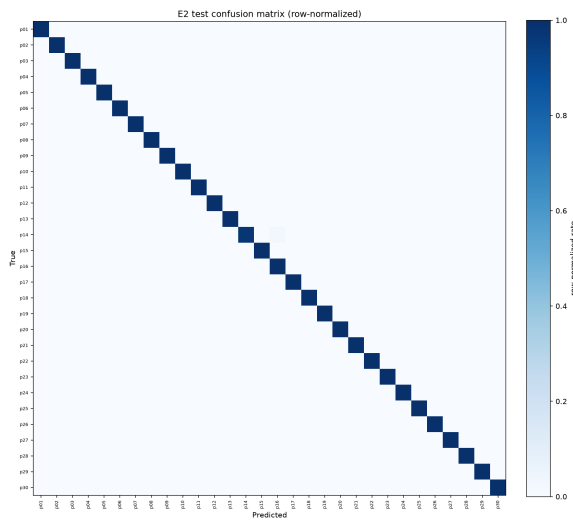
12. Deployment — Live Demonstration



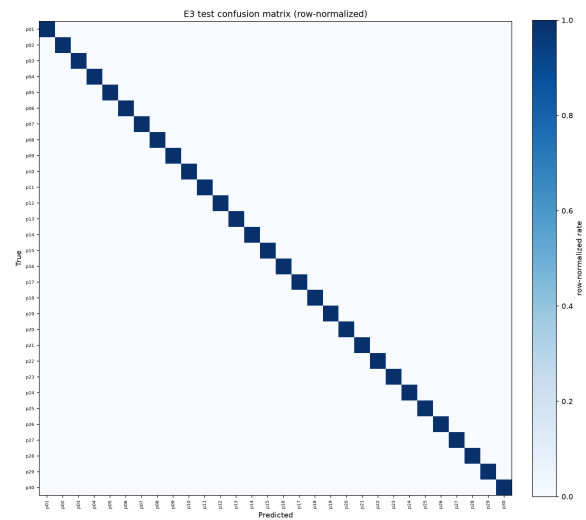
(a) E0 — nearest-centroid



(b) E1 — head, clean-only



(c) E2 — head, mixed-condition



(d) E3 — partial backbone fine-tune

Figure 1: Row-normalised 30×30 test confusion matrices for all four experiments. Darker cells indicate a higher share of that identity's test images landing in that predicted column; a perfect classifier is a solid diagonal.

Table 7: Approximate comparison with related published work. Numbers for related work are as reported in the cited papers; this project’s number is E3, this project’s own strongest result.

Work	Task	Scale	Reported masked/occluded accuracy
This project (E3)	Closed-set identification, 30 identities	5,988 images, 8 occlusion types	100.00% top-1 (998/998 test images correctly classified, 95% CI [99.6%, 100.0%])
Boutros et al. 2022 [3]	Open-set verification	Standard large face-verification benchmarks with masks added	Reports verification accuracy in the high-90s% on masked benchmarks, an improvement of several points over a plain ArcFace baseline
InsightFace MFR Challenge 2021 [4]	Open-set verification / identification	Millions of identities, web-scale	Top challenge entries report masked-face TAR in the 90s% at very low FAR; substantially harder open-set setting than this project’s closed set
NIST FRVT Part 6B 2020 [8]	Verification, commercial algorithms	Large government- operational dataset	Masked-face accuracy typically drops 5–50 percentage points relative to unmasked, depending on algorithm and mask type

12.1 Gallery and Threshold Calibration

The deployed system uses the gallery (nearest-centroid) approach rather than a fixed trained head, both because E0/E3 showed it is at least as accurate and because it allows a new occlusion example to be added to a participant’s centroid without retraining. Centroids are built from all 5,988 images (train+validation+test combined) for deployment, one 512-D L2-normalised mean embedding per identity. An UNKNOWN-rejection threshold is calibrated from the 1st percentile of genuine train-centroid-vs-held-out-validation cosine-similarity scores, giving a deployed threshold of 0.3179 — so that a face that does not match any enrolled participant well enough is reported as UNKNOWN rather than forced into one of the 30 identities.

12.2 Live Phone-Camera Demo

The demo runs as a Flask web server on a laptop, serving a browser page to a phone over the local Wi-Fi network via HTTPS (a self-signed certificate, generated once and reused across restarts). The phone’s camera streams frames to the server roughly every 400 ms via `getUserMedia`; each frame is detected, aligned with the identical `align_crop` function used in preprocessing, embedded, averaged over a short rolling buffer of recent frames for temporal stability, and matched against the gallery with the calibrated threshold. The result — participant name/ID, similarity score, and threshold — is rendered back on the phone screen in real time. A device/camera-lens picker is included so the operator can choose a sharper physical camera on phones with multiple rear lenses (e.g. avoiding an ultra-wide lens), and a front/back camera toggle is provided.

13. Deviations from the Proposal

Table 8: Deviations between the original proposal and the completed project.

Proposal said	What was actually done	Why
Three capture sessions (train/val/test each its own session)	One capture session per participant; group-aware stratified split used instead	Time constraints during collection allowed only one sitting per person
Accuracy reported by head pose (front/left/right/up/down)	Pose-level breakdown dropped	Pose was not separately labelled during capture
Conditions: mask+hat, generic “partial object”	Conditions actually collected: none, cap, mask, scarf, clear glasses, sunglasses, mask+sunglasses, mask+clear glasses	Reflects what was practically collectable; clear-glasses conditions were added instead
10–15 participants	30 participants	Six team members each recruited five people instead of a shared, smaller pool
From-scratch CNN baseline for comparison	Dropped; E0 (untrained nearest-centroid) used as the informative lower bound instead	E0 already showed the pretrained embedding was close to saturated, making a from-scratch CNN comparison less informative than the E1-vs-E2 occlusion-training comparison

The single-session change is the most consequential deviation: a single-session split cannot demonstrate generalisation across days, clothing, or setting the way a multi-session protocol would, and the reported accuracies should be read with that in mind (see Limitations, below).

14. Limitations and Future Work

- **Single session.** Train, validation, and test all come from one sitting per participant — same clothing, background, and day — which inflates accuracy relative to a genuine multi-session protocol.
- **Closed set only.** Every input is forced into one of 30 identities unless the deployed rejection threshold fires; no formal open-set evaluation (e.g. measuring false-accept rate against genuinely unenrolled people) was performed.
- **Camera confound.** Each team member photographed their own five people with one phone, so camera/device signature correlates with a specific group of five identities. Training-time augmentation (JPEG re-compression, sensor noise) mitigates but does not eliminate this.

- **429 exact-duplicate images, treated as independent.** Roughly 8% of test images are byte-identical to a training image; this is disclosed in Section 10 rather than corrected for, per the project’s decision to keep the dataset frozen once collection finished.
- **30 identities.** Results will not extrapolate to hundreds or thousands of enrolled people, and the closed-set numbers here are not comparable in absolute terms to the open-set, web-scale results of Section 11.
- **No anti-spoofing or liveness detection.** This is a laboratory classification prototype, not an access-control system, and would need a liveness check before any real deployment.
- **Future work** suggested directly by Section 3’s related work: a mask-aware triplet loss ([3]) or a restoration/de-occlusion front end ([5]) could be evaluated on a second, genuinely multi-session capture round to test whether either helps once the current dataset’s ceiling effect (E0 already at 99.5%) is removed by harder, cross-day, cross-location test data.

15. Conclusion

This project set out to build and evaluate a closed-set face-identification system robust to real occlusions, using a pretrained ArcFace model. Thirty participants were recruited and consented, and 6,000 images were collected across eight occlusion conditions and two lighting levels — exceeding the original scale target. A reproducible detect-align-crop pipeline processed 99.8% of the dataset successfully, and a duplicate-aware, coverage-guaranteed split produced a 3,546/1,444/998 train/validation/test partition. Four experiments, evaluated with full confusion matrices and per-class/per-occlusion precision, recall, and F1 (not only top-line accuracy), showed that a pretrained ArcFace embedding, used with no training at all, already reaches 99.60% test accuracy (E0); that training a classification head only on clean faces degrades specifically on the hardest combined occlusion, **mask+sunglasses** (E1, 94.29% overall, 71.4% on that condition), while training on occlusion-diverse data recovers full performance there and 99.90% overall (E2); and that partial backbone fine-tuning closes the remaining gap entirely (E3, 100.00%, perfect on every occlusion condition). The completed system was deployed as a live, phone-camera-driven demonstration accessible over the local network, using the gallery-plus-threshold approach validated as at least as strong as any trained head. Compared against three closely related, more sophisticated published approaches, this project’s simple pipeline performs surprisingly well precisely because its task — closed-set identification among 30 known people — is fundamentally easier than the open-set, web-scale problems those works address; the honest scientific contribution here is demonstrating how much of that gap a pretrained, off-the-shelf model closes on its own, and exactly where (occlusion-diverse training data, not further architecture or loss engineering) the remaining gap actually lived.

References

- [1] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” in *Proceedings of CVPR*, 2019.
- [2] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, “RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild,” in *Proceedings of CVPR*, 2020.
- [3] F. Boutros, N. Damer, F. Kirchbuchner, and A. Kuijper, “Self-restrained Triplet Loss for Accurate Masked Face Recognition,” *Pattern Recognition*, vol. 124, 108473, 2022.
- [4] J. Deng, J. Guo, X. An, Z. Zhu, and S. Zafeiriou, “Masked Face Recognition Challenge: The InsightFace Track Report,” in *Proceedings of ICCV Workshops*, 2021.

- [5] N. U. Din, K. Javed, S. Bae, and J. Yi, “A Novel GAN-Based Network for Unmasking of Masked Face,” *IEEE Access*, vol. 8, pp. 44276–44287, 2020.
- [6] Z. Wang, G. Wang, B. Huang, Z. Xiong, Q. Hong, H. Wu, P. Yi, K. Jiang, N. Wang, Y. Pei, H. Chen, Y. Miao, Z. Huang, and J. Liang, “Masked Face Recognition Dataset and Application,” *arXiv:2003.09093*, 2020.
- [7] C. Wang, H. Fang, Y. Zhong, and W. Deng, “MLFW: A Database for Face Recognition on Masked Faces,” in *Proceedings of CCBR*, 2021.
- [8] M. Ngan, P. Grother, and K. Hanaoka, “Ongoing Face Recognition Vendor Test (FRVT) Part 6B: Face Recognition Accuracy with Face Masks Using Post-COVID-19 Algorithms,” NIST Interagency Report 8331, 2020.